Singular Dynamic AMM Pricing Perpetual Options Across the Strike Continuum

Anonymous author
Anonymous affiliation

Abstract

Existing on-chain perpetual-option venues fragment liquidity across discrete strikes and rely on complex multi-pool mechanisms; pricing is rarely closed form, and the entire out-of-the-money continuum is never served from a single pool.

We present a market mechanism in which one liquidity pool, governed by the weighted constant-product invariant $(x^w) \cdot (y^{1-w}) = k$ with a dynamic weight w, prices the entire out-of-the-money strike continuum — both calls and puts — in closed form. Positions are barrier instruments. A position composed with the perpetual future it is opened against caps the trader's drawdown without forcing position closure, and settlement is denominated in units of that perpetual future. Two of the core results are formally verified in Lean 4 [6, 10]: (i) a composite-ray closed-form valuation identity that extends across the OTM-to-ITM boundary under an effective-strike substitution, and (ii) a no-internal-arbitrage characterisation as a biconditional — a costless collar yields zero capital-efficiency surplus if and only if the pool is symmetric — established by direct derivation with an explicit skew counterexample.

2012 ACM Subject Classification Theory of computation → Algorithmic mechanism design; Applied computing → Economics; Theory of computation → Program verification

Keywords and phrases automated market maker, perpetual options, barrier options, constant function market maker, decentralized finance, formal verification

Digital Object Identifier 10.4230/LIPIcs.AFT.2026.

1 Introduction

Perpetual futures markets have become the dominant venue for leveraged speculation on cryptocurrency prices, but a leveraged perpetual position carries the discontinuous risk of liquidation: when collateral runs out, the position is involuntarily closed at a loss. A natural hedge against this risk is an out-of-the-money (OTM) option on the same underlying, which caps the trader's drawdown past the strike. Existing on-chain venues that offer such options, however, fragment liquidity across discrete strikes and use complex multi-pool mechanisms; pricing is rarely closed form, and the entire OTM strike continuum is never served from a single pool.

This paper presents a market mechanism in which a single liquidity pool, governed by the weighted constant-product invariant $(x^w) \cdot (y^{1-w}) = k$ with a *dynamic* weight w, prices the entire OTM strike continuum — for both calls and puts — in closed form. The instruments are barrier options, and a position composed with the perpetual future it is opened against (its *origin perp*) caps the trader's drawdown without forcing position closure; the option does not expire, so the hedge persists for as long as the underlying perp is held. Settlement is denominated in units of the carved-perp slice that the band escrows against, with a single dollar-conversion at close.

Contributions.

- A closed-form pricing mechanism for the entire OTM perpetual-option continuum (both calls and puts) from a single dynamic-weight CFMM pool, with the weight updating on every trade according to a conservation law on $\alpha = xw$ and $\beta = y(1-w)$.

- A geometric framing in which strikes are identified with rays in the (x, y) plane, valuation and funding are read off intersections of those rays with two distinct curves (a live pool curve and a fixed anchor curve), and the close-equals-exercise mechanism is realised as a ray-parking property at the pool's at-the-money line.
- A two-jobs settlement protocol that values each leg of a band in carved-perp units and converts to dollars with a single equity multiplication, with a dimensional-check argument confirming the conversion is balanced at the level of the formula.
- A no-internal-arbitrage characterisation as a genuine biconditional: a costless collar yields zero capital-efficiency surplus if and only if the pool is symmetric ($w = \frac{1}{2}$). The skew direction is established by an explicit counterexample.
- Two of the core results — the composite-ray closed form across the OTM-to-ITM boundary, and the no-arbitrage biconditional — are formally verified in Lean 4 [6, 10], with no domain-specific axioms.

Organization.

Section 2 defines the instruments and their pricing. Section 3 gives the curve-skewing intuition behind the AMM construction. Section 4 sets up the strike-as-ray translation. Section 5 states the conservation law and trade formula. Sections 6 and 7 cover strike normalisation and funding. Section 8 introduces the origin perp; Sections 9 and 10 cover the position life-cycle and settlement, including the explicit closed-form payout formula with its dimensional check. Section 11 discusses liquidity provision, Section 12 the no-arbitrage and round-trip properties, Section 13 limitations, Section 15 related work, and Section 14 concludes. Appendices contain the notation table, derivations, identity statements, the three-operator structural lemma, the unit-chain discipline, the state-space geometry results, the Lean verification ledger, and a worked numerical example.

2 Perpetual Options

2.1 Definition

Perpetual options are non-expiring contracts-for-difference, representing the payout from favourable price action — upward for calls, downward for puts — of an underlying asset relative to a denominating asset (here, USD) beyond a given strike price. The instruments priced here are *barrier* options: a position's value is bounded, topping out at one unit of the underlying perpetual future when the position is in-the-money, and remaining at that ceiling as the position moves further in-the-money. A position is terminated when its holder closes it.

2.2 Utility

Liquidation in a perpetual future is the involuntary termination of a leveraged position when collateral is exhausted. Composing a perpetual future with an OTM perpetual option on the same underlying — a call to hedge a short, a put to hedge a long — caps realised loss past the strike, bounding the trader's drawdown without forcing position closure. Because the option position does not expire, the hedge persists for as long as the underlying perp is held.

2.3 Pricing

For an OTM strike, the value of a position is given by its **mark**: the ratio of the strike ray's slope to the pool's at-the-money mark, taken as the lesser of that ratio and its reciprocal,

$$\text{mark} \;=\; \min(\text{slope},\ 1/\text{slope})$$

so that the mark is a dimensionless quantity bounded in $(0, 1]$. (Appendix A provides a notation table the reader may consult on first symbol use.) Intuitively, the mark is the *fraction of a full perpetual future* that the position is worth: it approaches 1 as the position nears the money and decays toward 0 deep out-of-the-money. The value of a position of size q at a strike is $q \cdot \text{mark}$.

This pricing carries no separate premium object. The mark is the complete description of what a position is worth; there is no extrinsic-versus-intrinsic decomposition and no premium term layered on top. Throughout this paper, "the value of a position" means $q \cdot \text{mark}$, and nothing else.

3 AMM Intuition

We make two generalisations to the constant-function AMM $(x^w) \cdot (y^{1-w}) = k$, giving it the expressivity to price perpetual options across the strike continuum.

"Trades skew the AMM curve instead of moving the reserves point along it."

A two-asset constant-function AMM (CFMM) [1, 4] can be visualised as a curve in the (x, y) plane, along which the reserves (x, y) are a single point. Price is the slope at that point, and a trade moves the reserves point along the curve.

Two curves are in play, and it is worth naming both at the outset. The first is the **pool curve** $(x^w) \cdot (y^{1-w}) = k$, whose *shape* is set by the weight w; this is the curve that skews under trades, and the curve against which strikes are priced. The second is the **trajectory hyperbola** $(x - \alpha) \cdot (y - \beta) = \alpha\beta$, the locus along which the reserves point actually moves (derived in Section 5). The two are tangent at the reserves point — they share a slope there — which is why pricing read off the pool curve is faithful, and why "reshaping the curve" and "moving along the trajectory" describe the same event from two angles.

The pool curve gives a family of shapes indexed by w. Rather than updating the pool's quoted prices by moving the reserves point along a fixed curve, we obtain the same effect by changing the *shape* of the pool curve via w. For a small trade, if the reserves point were to move along the curve, we note the slope at the post-trade point; then, instead of moving the reserves point, we reshape the curve — by updating x, y, and w — so that the slope of that post-trade point is brought to the pre-trade reserves point. Section 5 makes this precise.

"Trades happen anywhere on the curve, not just at a single point."

In traditional CFMMs, trades are spot swaps, always occurring *at* the reserves point on the curve. We treat spot swaps as a special case of perpetual-option swaps, which may occur at *any* point on the curve, subject to that point being out-of-the-money. A transaction at any *trade point* on the curve is treated as if that trade point were the reserves point.

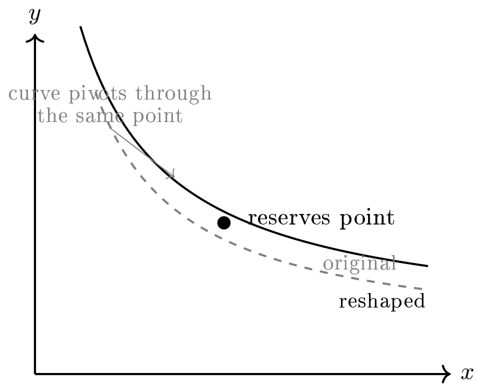

Figure 1 The pool curve reshapes around the reserves point via the weight w. The reserves point itself does not move; the pre- (dashed) and post-trade (solid) curves both pass through it, and the post-trade slope at the reserves point is the trade's marginal price.

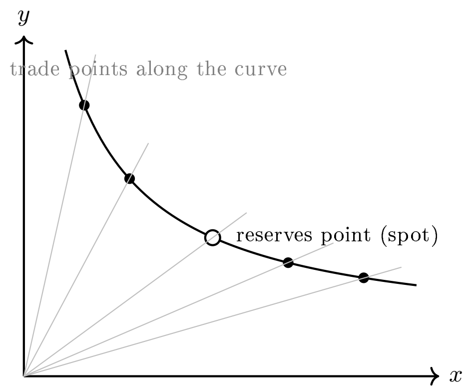

Figure 2 The reserves point (hollow circle) is the special case of a spot swap. Other trade points along the curve (filled) correspond to OTM transactions at different strike rays. Each ray from the origin intersects the curve once; that intersection is the trade point for the strike identified with the ray.

4 Translating Perpetual Options Into AMM Swap Transactions

Here we establish the conventions that let us translate perpetual-option transactions into the language of the AMM.

4.1 Mapping Strikes to Ray Angles

Each ray from the origin in the (x, y) plane is parameterised by its angle — equivalently, by the constant ratio y/x along the ray. We identify each ray with a unique perpetual-option strike. A ray intersects the pool curve at exactly one point; that intersection is the *trade point* for the strike.

4.2 The Anchor Curve and the Strike-to-Ray Map

The protocol uses a fixed reference object — the **anchor curve** — alongside the live pool curve. The anchor curve is the symmetric $w = \frac{1}{2}$ constant-product curve, whose at-the-money point coincides with the current oracle. As the oracle moves, the anchor moves with it, but the anchor never skews; its shape is fixed.

A strike has a fixed dollar identity, set at the moment it is opened. Its ray, however, is a *live* quantity, derived from the current oracle frame:

$$\theta(K, t) \;=\; K \,/\, \mathrm{oracle}(t)$$

normalised against the *live* oracle, not against an entry-frozen oracle. The ray re-derives from live state at every read; no ray is stored. Stability is at the dollar level — the strike's K is locked — not at the ray level. The ray moves to express where the frozen dollar strike sits, geometrically, right now.

The regime test for the strike — whether it is in- or out-of-the-money — is then a comparison between this ratio and the pool's live normalised marginal price s_{Norm}, both evaluated in the same frame. The pool's role enters through s_{Norm} in the regime test, not through the ratio itself; the ratio is what tells the regime test where to look.

Both trades and rebases shift strike rays. Trades shift them by moving the reserves point along the pool curve. Rebases shift them by rescaling the oracle frame ($\mathrm{oracle} \to r \cdot \mathrm{oracle}$). Either way, the dollar strike K stays locked; only the live ratio moves. There is no caching of strike rays across operations.

4.3 The 45° Ray Represents Spot

By convention, the 45° ray ($y = x$) corresponds to the at-the-money strike. The anchor curve is normalised so that its 45° intersection sits at unity (see Section 6). Rays on either side of 45° parametrise the OTM continuum.

4.4 Calls and Puts on Opposite Wings

Rays steeper than 45° and rays shallower than 45° partition the OTM continuum into two wings — one carrying calls (above-spot), the other puts (below-spot). The pool prices both wings from the same (x, y, w) state; no liquidity is partitioned between them.

4.5 Barrier as the Trade Primitive

The primitive transaction is a *barrier*: a position at a single strike ray, valued at $q \cdot \mathrm{mark}$. A *spread* — a position bounded between an inner and an outer strike on the same wing — is a composition of barrier primitives, valued as $q \cdot (\mathrm{mark}_{\mathrm{inner}} - \mathrm{mark}_{\mathrm{outer}})$. A *band* — sold and bought legs on opposite wings — is a further composition. Because the legs sit on opposite wings, a band cannot be reduced to a single-ray transaction, and is always treated as two legs (see Appendix C).

The same composition admits a closed-form shortcut for the *execution* of a spread, expressing it as a single AMM swap at a composite ray $\theta^* = \sqrt{\theta_{\mathrm{inner}} \cdot \theta_{\mathrm{outer}}}$ with half-spread $\delta = \frac{1}{2} \log(\theta_{\mathrm{outer}} / \theta_{\mathrm{inner}})$. The shortcut and the barrier-by-barrier expression are provably equal; we use the barrier-by-barrier form for valuation throughout the body, and the composite-ray shortcut for execution. Both are stated in Appendix C.

5 AMM Mechanics

The AMM accounts for every transaction as a swap of cash against the underlying at a trade point.

5.1 Conservation Law

For each trade, the quantities $x \cdot w$ and $y \cdot (1-w)$ are individually conserved. Denote $\alpha = x \cdot w$ and $\beta = y \cdot (1-w)$. The pool state is therefore fully determined by (x, y); the weight w is a derived field, $w = \alpha/x$ (equivalently $1 - w = \beta/y$), recovered from the conserved quantities at any time. No additional state storage is required.

Together with the constraint $w + (1-w) = 1$, the conservation law confines all reachable states to the hyperbola $(x-\alpha) \cdot (y-\beta) = \alpha\beta$ (see Appendix B). The invariant $k = x^w \cdot y^{1-w}$ is a *label* on the pool curve's level set — a dependent readout, not a primitive of pool state and not, on its own, a measure of liquidity depth.

5.2 Trade Formula

Parametrising a swap by its cash leg Δy — the cash into the pool, signed — the global state updates along the conservation hyperbola:

$$
\begin{aligned}
y' &= y + \Delta y \\
\Delta x &= -\,\alpha\beta \cdot \Delta y \,/\, [(y-\beta)(y'-\beta)] \\
\Delta w &= \beta \cdot \Delta y \,/\, [y \cdot y']
\end{aligned}
$$

The post-trade state is $(x', y', w') = (x + \Delta x,\, y + \Delta y,\, w + \Delta w)$, with $\alpha = x' \cdot w'$ and $\beta = y' \cdot (1 - w')$ preserved by construction.

The third update, Δw, is not an independent equation. Since $w = \alpha/x$ and α is conserved, Δw is fully determined by Δx; the expression above is the algebraic simplification of $\Delta w = \alpha/x' - \alpha/x$. It is written out because the weight w is the quantity the pool's pricing actually moves, and the reader is owed an explicit sight of its update.

The direction of the cash leg is not free. For a transaction at a given strike, whether cash flows into or out of the pool is fixed jointly by the wing and the side: buying a call and selling a put draw cash out of the pool; selling a call and buying a put pay cash in. Writing the wing factor as $+1$ for a call and -1 for a put, and the side factor as $+1$ for a sell and -1 for a buy, the sign of the cash leg is their product,

$$\operatorname{sign}(\Delta y) \;=\; (\text{call}\,?\,+1 \,:\, -1) \cdot (\text{sell}\,?\,+1 \,:\, -1).$$

This is what makes the two legs of a band — a sold leg on one wing, a bought leg on the other — transact in the *same* direction against the curve rather than cancelling, a fact Section 12.1 relies on.

Notional is range-invariant.

The size q of a leg is a fixed property of the leg. Adjusting a position's strike range — moving its inner or outer bound — reshapes the leg's value and its position on the curve, but does not change q. The range-sensitive quantities are the value and the cash leg of the swap; the notional is not among them.

5.3 Pool Depth Evolution

Pool depth at any ray is determined by (x, y, w) at that ray's trade point. As w updates with trades, depth redistributes smoothly across the continuum.

The invariant k is not, on its own, a measure of pool depth. Under strike normalisation (Section 6) a rebase multiplies k by a factor r^w, so k is not invariant across rebases and two pool states cannot be compared by their bare k values. Depth is expressed by the rebase-aware relation in which k sits, not by k in isolation.

6 Strike Normalisation

6.1 Rationale

Sustained directional drift in spot causes cumulative reserve imbalance — one wing depletes faster than the other. Without normalisation, the strike-to-ray map drifts away from its anchor, and the live pool decouples from the oracle. Strike normalisation keeps the pool's pricing frame anchored as spot moves, so that the strike-to-ray identification remains stable under arbitrary spot trajectories.

6.2 Approach

We index the oracle spot price to unity and define the anchor curve in this normalised frame. As oracle spot moves, the frame is periodically *rebased*: the reserves and conserved quantities are rescaled by a factor r,

$$x \to r \cdot x, \quad \alpha \to r \cdot \alpha, \quad k \to r^w \cdot k$$

with the rebase carrying the pool's frame along with the traveling spot. The pool's at-the-money mark floats around the unity reference.

Two points are worth making explicit. First, a rebase **bounds drift; it does not nullify the pool's deviation from the anchor curve**. A rebase prevents directional drift from accumulating without limit, but the pool may legitimately sit at a deviation from the anchor at any given time — that standing deviation is precisely what the funding mechanism responds to (Section 7). Second, **trades and rebases commute on final state**: applying a trade and then a rebase yields the same (x, y, w) as applying the rebase and then the trade. They do not, however, commute on *slippage* — a rebase moves the reserves point along the curve, so a subsequent trade sweeps a different arc than it would have absent the rebase, and the slippage paid is different. Round-trip neutrality therefore holds only in a fixed frame.

7 Funding

7.1 Intuition

Funding anchors the pool's pricing to the oracle. By analogy with a perpetual future — where funding flows according to the sign of the mark-to-index basis — a perpetual-option position accrues funding according to how the pool prices it relative to its anchor reference. As a structural consequence, funding is countercyclical: traders on the crowded, consensus side of the market pay traders on the contrarian side, dampening the variance of LP exposure.

7.2 Funding for a Given Position

Funding for a position is determined by the deviation of the live pool curve from the anchor curve, evaluated **at that position's own strike ray**.

The *magnitude* of funding is the size of this pool-versus-anchor deviation at the position's ray. The *direction* follows the perpetual-future analogy: if the pool prices the option above its anchor-curve value — its mark exceeds the value the anchor curve assigns at that ray

— then the long side pays funding; if below, the long side receives it. Because deviation is evaluated per ray, funding is naturally a per-position quantity, and the signed funding of each leg of a multi-leg position follows directly from the deviation at each leg's ray.

A small but important note: the anchor curve, not just the oracle, is what enables per-strike funding. The oracle supplies a single number — one value at one ray (the ATM ray). The anchor curve supplies a deviation at every ray, which is what a per-strike funding quantity requires.

The frame for funding is the **pool-versus-anchor deviation at the position's own ray** — not pool spot, and not the pool's shape considered globally. A rebase does not change this quantity directly; it bounds the rate at which directional drift accumulates into it. Funding prices the standing deviation; rebasing caps how large that deviation can grow. Stated this way the two mechanisms are consistent: each acts on the same per-ray deviation, funding by responding to its level and rebasing by limiting its accumulation.

8 The Origin Perp

A perpetual-option position in this protocol is never opened in isolation. It is opened as a *band* — a composition of option legs — against a specific perpetual future the trader already holds. The band exists to protect that perp from liquidation, and everything about how the band is valued and settled refers back to it. We call it the **origin perp**.

A trader's perpetual-future exposure is held as a single aggregated position, not as a stack of discrete lots. Adding size in the same direction grows the one position; reducing it shrinks it. There is one perp, with one notional and one equity, moving continuously as the trader trades and as the mark moves.

When a band is opened, it is minted against a **carved slice** of that aggregated perp — a defined portion, not the whole. At the moment of opening, three quantities are recorded and thereafter held immutable for the life of the band: the slice's absolute notional, its equity, and the perp's mark at the moment of entry. Later activity on the aggregated perp — more size added, the mark moving, equity changing for unrelated reasons — does not touch the slice. The band carries its own origin slice as a fixed reference.

This freezing is what keeps settlement well-posed. Because the slice is fixed in absolute terms, the band's settlement arithmetic cannot be contaminated by perp activity unrelated to the band. The perp P&L attributable to a band is simply the P&L of its frozen slice, tracked forward from entry — not a re-derived share of a live, moving aggregate. The slice is the band's immutable record of what it was opened against.

9 Position Lifecycle

A perpetual-option position has three phases: open, hold, close.

9.1 Open

A position is opened as a band, carved against an origin perp (see The Origin Perp). The trader selects strikes on the OTM continuum — subject to the OTM restriction — and the band's legs are recorded against their trade points, with the AMM reshaping via w and preserving the conservation law at each leg.

Positions are opened as **premium-neutral structures**. A band's legs are arranged so that their cash legs offset: the cash raised by the sold leg funds the cash spent on the bought leg, and no net cash crosses the boundary between trader and pool on opening. There is no bare cash-for-option trade exposed to the trader.

It is worth distinguishing two layers here. At the level of AMM accounting, each leg is still a swap of cash against the underlying at a trade point — this is how the pool records the transaction, and the Trade Formula applies leg by leg. But that cash leg is an accounting quantity internal to the pool's bookkeeping. At the trader-facing level, a position is a premium-neutral structure.

Three quantities are frozen at the moment a band is opened, and together they form a recurring design pattern across the protocol: the dollar strikes of the legs, the carved origin slice (its notional, its equity, and the perp's mark at entry), and the leverage L_0. Every quantity that matters at close is either derived live from current state or read off one of these three frozen records. This is what makes settlement attribution well-posed across the band's entire life.

9.2 Hold

An open position accrues funding continuously, as a function of the deviation between the live pool curve and the anchor curve at the position's ray (Section 7). The position carries no expiry.

9.3 Close

The holder closes a position by reversing it on the AMM at its strike rays. The closing transaction is not subject to the OTM restriction — that restriction governs opening only — so a position whose strike has been crossed by the pool's at-the-money mark, and is now in-the-money, is closed through the same mechanism.

Closing and exercising are economically equivalent for the holder, and the lookup geometry is what makes this so. A strike's ray expresses where its frozen dollar K sits geometrically in the live frame: the ratio $\theta(K, t) = K/\text{oracle}(t)$ moves as both trades and rebases move the pool curve. While the leg is OTM, the ray sits on the OTM side of the pool's at-the-money line and moves with the live frame. When the pool's marginal price crosses the strike — that is, when s_{Norm} crosses θ — the leg goes in-the-money, and its ray **parks at the pool's at-the-money line**: it does not cross. Mathematically, the parking is realised by the leg's mark saturating to 1; the ray clamped at ATM and the mark capped at unity are the same condition viewed two ways. While the leg is ITM, its effective strike is the pool's at-the-money mark itself, and the ray holds parked there however far the pool pushes. If the pool later reverses, the ray un-parks and slides back out onto its own wing.

This geometric picture *is* the close-equals-exercise mechanism. When a leg is closed in-the-money, the closing swap is valued at its effective strike — the pool's at-the-money mark — and that valuation returns the leg's full intrinsic value naturally, with no separate exercise path. The composite-ray closed form for settlement is verified to hold across the OTM-to-ITM boundary under this effective-strike substitution (result C1; see Appendix G).

The parking mechanism handles the ordinary trajectory case: a strike's ray that would otherwise cross to the other wing instead parks at the at-the-money line. An additional live **wing-lock guard** runs at each AMM-reversal swap as a check against adversarial cases — a mis-tagged leg, or a swap-direction error that would put the transaction on the wrong wing relative to the pool's live mode ray. The two mechanisms are complementary: parking handles ordinary state evolution, wing-lock catches manipulation.

A band is closed in a single composite-ray operation at the AMM execution level. Its legs are clubbed and the band is settled as one transaction rather than unwound piece by piece; the composite-ray construction used to open a spread extends to closing a whole band.

10 Settlement

10.1 The Escrow Frame

It is useful to approach settlement through a familiar idea first. A leveraged perpetual future escrows **margin**: the trader posts collateral, and the position settles against that posted margin — gains and losses are reckoned against it, and what remains is returned on close. This is the standard picture.

This protocol adds one further layer of the same idea. Opening a band escrows a **slice of the perp itself** — the carved origin slice of the previous section — and the band settles in terms of that escrowed perp, just as the perp settles in terms of escrowed margin. The band is a claim that resolves against escrowed perp the way a perp is a claim that resolves against escrowed margin. Settlement is therefore **perp-denominated**: a band's leg values are carried in units of the carved perp, not as standalone cash amounts.

One consequence follows immediately. What is escrowed is a *live* slice of a perp, not a frozen sum of cash. Settling in terms of it therefore means settling against its **equity** — its entry margin adjusted by the P&L attributable to it — at the time of close. The leg values, expressed in carved-perp units, are converted to cash at the carved slice's **closing equity**. This conversion is not an adjustment bolted onto settlement; it is the honest exchange rate between carved-perp units and cash at the moment of close. It deflates the result when the slice has lost value and inflates it when the slice has gained, symmetrically, because that is what the slice is worth.

10.2 Closing a Band

A band has two legs on opposite wings — one above spot, one below. Because spot cannot be in-the-money on both wings at once, **at most one leg of a band is in-the-money at any time**. Settlement therefore has a clean two-case structure.

When no leg is in-the-money, both legs are live positions on the pool curve, and both are closed by reversing them on the AMM in the usual way.

When one leg is in-the-money, that leg has value but no live curve position to reverse — it is **settled to cash directly**, its value given by the composite-ray closed form under the effective-strike substitution (result C1; see Appendix G). The remaining out-of-the-money leg is live and is reversed on the AMM normally. In this case, sequencing matters: value the settled-to-cash leg before any AMM swap perturbs the pool, so that the in-the-money leg is priced against the pool state as it was at the moment of close.

10.3 The Settlement Formula

Let X denote the sold leg's settled value and Y the bought leg's, both in carved-perp units, with $\text{raw\_net} = Y - X$. The carved slice's closing equity is computed from its frozen origin record and the perp's mark at close:

$$\text{attributablePnL} = \text{carvedNotional} \cdot \frac{\text{perpMark}_{\text{now}} - \text{entryPerpMark}}{\text{entryPerpMark}}$$

$$\text{carvedEquityAtClosure} = \text{carvedEntryEquity} + \text{attributablePnL}$$

Settlement then performs the unit conversion to dollars and the leverage amplification in a single tail multiplication:

$$\text{trader\_payout} = L_0 \cdot \text{raw\_net} \cdot \text{carvedEquityAtClosure}$$
$$\text{club\_delta} = (L_0 - 1) \cdot \text{raw\_net} \cdot \text{carvedEquityAtClosure}$$

with a counterparty-equity floor as final guard: if raw_net > 0 and the counterparty club's equity is non-positive, both trader_payout and club_delta are zero.

The / entryPerpMark normalisation in attributablePnL is dimensionally required. The carved notional is in dollars, the difference of perp-marks is in dollars-per-asset, and the division turns that difference into a dimensionless fractional move. Without the division, the product would have units of [USD^2/asset] — a unit that does not exist. With it, attributablePnL is in dollars, and adds cleanly to carvedEntryEquity (also in dollars). The perp's mark is sourced from the perp-DEX feed in production; in the simplified single-venue setting we collapse it numerically to the oracle, which does not change the formula's shape.

10.4 Two Scalars

Two distinct quantities act at settlement, and they must not be conflated. The first is the carved slice's **closing equity**, which performs the unit conversion from carved-perp units to cash. The second is L_0, the leverage frozen at band-open, which amplifies the settled net. One sets the exchange rate; the other sets the magnification. They apply at different points: the net is computed in carved-perp units, converted to cash at the closing equity, and then amplified by L_0, with the counterparty equity floor applying as a final guard.

11 Liquidity Provision

Liquidity providers deposit proportional contributions of cash and underlying into the pool reserves (x, y), in the manner of a weighted constant-product pool [8], and receive pool-share tokens representing a claim on (x, y) at any future state. Standard CFMM-LP constructions [1, 2] apply; we do not elaborate them here. Liquidity provision may be levered synthetically — the LP takes on more pool exposure than they deposit, using the same leverage architecture that powers perps and bands — though the implementation specifics are out of scope for this paper.

LPs are collectively the counterparty to every open option position in the pool. Net LP exposure at any moment is the position-weighted aggregate of option deltas across all open positions on both wings.

LP P&L from internal protocol accounting equals fees earned. Reserve value changes are an impermanent-loss / HODL-benchmark concern — an external analysis frame — and are not part of protocol accounting; including them would double-count against fee P&L.

11.1 Hedging Note

LPs can hedge net delta externally via spot or perpetual futures on the same underlying. The countercyclical funding mechanism partially offsets directional drift in LP exposure — funding inflows are systematically aligned with the wing under stress — but does not fully neutralise it. The residual is the LP's structural option-writer position.

12 Properties

12.1 No Internal Arbitrage

Within a single pool state (x, y, w), no composition of swaps across rays yields a riskless capital-efficiency edge. The sharp statement concerns the **costless collar**: a band whose sold and bought legs are arranged to be cash-neutral by construction.

The collar yields **zero capital-efficiency surplus if and only if the pool is symmetric** — that is, if and only if the weight $w = \frac{1}{2}$. Under skew, $w \neq \frac{1}{2}$, the surplus is non-zero, with an explicit counterexample. The "if and only if" is the substance of the claim: the result discriminates between symmetric and skewed pools, which is the evidence that it is structural rather than an artifact of the definitions. The capital-efficiency edge of a costless collar is thus a *skew phenomenon* — it exists only when the pool is directionally skewed, and vanishes exactly at symmetry. This result is verified in Lean 4 (see Appendix G).

A distinction must be drawn carefully. The no-arbitrage result is a statement about **pricing** — it says that at symmetry there is no per-unit pricing edge to extract. It does **not** say the pool is left unchanged by a costless collar. A costless collar costs zero net cash, but its two legs transact in the same direction against the curve, so the pool does move — the weight w shifts. The trader pays for that displacement: the proceeds of a swap are the path integral of the marginal price along the curve [3], so any displacement of the pool is internalised, by construction, by the trader who causes it. The legs therefore transact at fair, sequenced prices, and that price path is the cost of moving the pool. A collar can therefore cost zero cash and still do real work on the pool, with no contradiction — because the displacement was paid for, the question "who pays for the arbitrage?" has no answer, and hence there is no arbitrage. In particular, a collar opened at symmetry may leave the pool skewed; that induced skew is simply pool state, bought and paid for at fair prices, and the next trader transacts fairly against it.

The argument given here is intuition; the formal proof of the biconditional is the Lean result C4 in Appendix G, which derives the surplus identity from the concrete $\min(\text{slope}, 1/\text{slope})$ mark formula and produces an explicit skew counterexample for the negative direction.

The slippage point sharpens further. Because slippage is the path integral of marginal price along the swept arc, and the arc depends on where the reserves point starts, a rebase that moves the reserves point changes the slippage a subsequent trade would pay. Round-trip neutrality therefore holds only within a fixed frame.

This is a statement about *free cash*. Liquidation protection retains genuine risk-management value even at zero skew: a leveraged perp faces a discontinuous, ruinous liquidation, and capping smooth upside is not symmetric in utility with preventing a wipeout. Zero skew removes the *free* capital-efficiency edge, not the *worth* of the protection.

Finally, the collar's capital-efficiency edge and the protocol's funding flow are the same underlying quantity — directional imbalance — observed twice: no skew means no edge and no funding flow, and the two move together.

12.2 External Arbitrage

The pool's at-the-money mark can diverge from the oracle between blocks; arbitrageurs restore convergence by trading against the pool. This is the price-discovery mechanism, incentivised by the funding-elastic restoring force (Section 6).

12.3 Round Trip

Opening and immediately closing a position at the same rays, in the same frame, returns the original position to within fees. Cross-frame round trips are not neutral, for the slippage-along-the-arc reason given in Section 12.1.

13 Limitations

The design is presented for a single liquidity pool. A single pool exposes one directional weight w and one rebasing frame; it is a deliberately rigid pricing system, with no per-strike liquidity-provider discretion. This rigidity is the source of the closed-form tractability and the structural no-arbitrage property, but it also fixes the shape of the pool's liquidity distribution: the protocol prices coherently within one fixed family of pricing surfaces. The calibration of funding elasticity and related parameters is protocol-specific and left to implementation.

Several further limitations are worth flagging explicitly.

Oracle dependence.

The strike-to-ray map is normalised against a live oracle, and in the present MVP-level treatment the perpetual-future's mark is collapsed numerically to the same oracle. Production deployment requires a robust oracle architecture (e.g. multi-source aggregation with manipulation-resistant medians), and the perp-mark feed in the settlement formula must be sourced from the actual perp-DEX feed rather than the oracle, to handle DEX-vs-oracle basis correctly. Adversarial oracle manipulation is not analysed in this paper and would interact non-trivially with the funding and rebasing mechanisms.

Single-asset, single-pool scope.

Multi-asset extensions of the weighted constant-product family are well-known [8], but the strike-to-ray translation and the settlement protocol developed here are presented for a single underlying. Adapting the closed-form valuations to a multi-asset basket would require, at minimum, a careful generalisation of the wing-and-ray structure and is left for future work.

No empirical or simulation validation.

The results in this paper are theoretical (closed-form derivations) and formal (Lean-verified identities). The paper does not include simulation studies of pool behaviour under realistic price trajectories, sensitivity analyses on funding-elasticity parameter choices, or empirical comparisons against existing perpetual-option venues. Such studies are natural next steps and would strengthen claims about practical deployability.

Extreme-trajectory behaviour.

The pool curve's depth distribution is fixed by the constant-product form; behaviour under sustained directional spot trajectories (where rebasing happens frequently) and under sharp discontinuous moves (where many positions cross from OTM to ITM in close succession) is bounded by the rebase-aware framework of Section 6 but not characterised quantitatively. Implementations should perform stress-testing against historical extreme-move scenarios before production deployment.

Security audit and deployment.

The mechanism described in this paper has not been independently security-audited as a smart-contract implementation. Any deployment to a live blockchain environment would require independent code review, formal verification of the on-chain implementation against the specification here, and standard DeFi-protocol security practices (re-entrancy guards, integer-overflow handling, upgrade paths, time-locked governance). The formal-verification results presented in this paper concern the mathematical mechanism, not any specific implementation.

14 Conclusion

A single liquidity pool governed by a weighted constant-product invariant with one dynamic weight prices the entire OTM perpetual-option continuum in closed form. Positions are barrier instruments, priced by a bounded mark function and carrying no separate premium object. Composition of a position with the perpetual future it is opened against — its origin perp — delivers the liquidation protection that motivates the design, and settlement is denominated in units of that perp, with the carved origin slice keeping settlement attribution well-posed.

Three quantities are frozen at the moment a band is opened — the dollar strikes, the carved origin slice, and the leverage — and together they make settlement attribution well-posed across the band's life. The pool's no-internal-arbitrage property is structural: a costless collar yields zero capital-efficiency surplus if and only if the pool is symmetric, a biconditional verified in Lean 4.

15 Related Work and Acknowledgements

The weighted constant-product invariant $(x^w)\cdot(y^{1-w}) = k$ that underlies the present pricing surface was introduced by Balancer [8] as a generalisation of the constant-product invariant of Uniswap, with the weight w *static* — fixed at pool creation and never updated. The present design takes the same invariant and makes the weight *dynamic* — w updates on every trade according to the conservation law of Section 5, with $\alpha = xw$ and $\beta = y(1-w)$ preserved through each trade. The use of dynamic weights on the weighted constant-product invariant has been independently explored by QuantAMM [9] as a mechanism for time-varying portfolio rebalancing; we acknowledge their concurrent work on dynamic-weight CFMM mechanics. The application here — pricing perpetual options across the strike continuum via per-trade weight updates — is, to our knowledge, novel.

The continuous-function market maker framework of Angeris and colleagues [3, 4] provides the foundational analytical lens (curve as pricing surface, swap as marginal-price path integral) used throughout this paper.

The barrier-instrument framing for non-expiring options on automated market makers was developed concurrently in several venues. The concept of non-expiring options with continuous funding payments was introduced by White and Bankman-Fried [11] ("everlasting options"); a closely related construction is Panoptic [7], which prices non-expiring options on Uniswap v3 LP positions. The present construction differs in unifying the entire OTM continuum — both wings, all strikes — within a single dynamic-weight pool, rather than distributing positions across a tick-grid or constructing payoffs from concentrated-liquidity primitives.

A note on drafting.

Large language models were used as drafting and editing assistants in the preparation of this paper. The technical content — the protocol design, the formulas, the Lean 4 verification, and all design decisions — is the authors' own. The models served as a writing and structuring aid; every claim in this paper has been verified by the authors against the underlying formal specification and reference implementation. We mention this in the interest of transparency.

References

1. Hayden Adams, Noah Zinsmeister, and Dan Robinson. Uniswap v2 core. Whitepaper, 2020.
2. Hayden Adams, Noah Zinsmeister, Moody Salem, River Keefer, and Dan Robinson. Uniswap v3 core. Whitepaper, 2021.
3. Guillermo Angeris and Tarun Chitra. Improved price oracles: Constant function market makers. In *Proceedings of the 2nd ACM Conference on Advances in Financial Technologies (AFT '20)*, 2020.
4. Guillermo Angeris, Hsien-Tang Kao, Rei Chiang, Charlie Noyes, and Tarun Chitra. An analysis of Uniswap markets. Cryptoeconomic Systems, 2019.
5. Anonymous. Formal verification artifacts for the present paper (Lean 4). Available from the authors on request; will be deposited in a public repository upon publication, 2026.
6. Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *Automated Deduction – CADE 28*, 2021.
7. Guillaume Lambert, Jesper White, Dan Robinson, and Jean-Marc Bordignon. Panoptic: A perpetual, oracle-free options protocol. Whitepaper, 2023.
8. Fernando Martinelli and Nikolai Mushegian. Balancer: A non-custodial portfolio manager, liquidity provider, and price sensor. Whitepaper, 2019.
9. QuantAMM. Temporal-function market making and dynamic-weight constant-function market making. Project documentation, 2024.
10. The mathlib Community. The Lean mathematical library. In *Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP '20)*, 2020.
11. Dave White and Sam Bankman-Fried. Everlasting options. Paradigm Research, 2021.

A Notation Table

Symbol	Meaning
x, y	Pool reserves of the two assets (underlying, cash)
w	Dynamic weight; $w \in (0, 1)$
k	Pool curve label; $k = x^w \cdot y^{1-w}$ (not a depth measure)
α	Conserved quantity; $\alpha = x \cdot w$
β	Conserved quantity; $\beta = y \cdot (1 - w)$
(x_T, y_T)	Trade point on the pool curve
θ	Ray angle from origin; identified with a strike
K	Strike price (dollar, frozen at open)
oracle(t)	Live oracle price; defines anchor curve's ATM
s_{Norm}	Pool's normalised marginal price; $s_{\text{Norm}} = (1 - w)/w$
q	Position size (notional) of a leg; range-invariant
mark	Position value fraction; $\min(\text{slope}, 1/\text{slope}) \in (0, 1]$
θ^*	Composite ray of a spread; $\theta^* = \sqrt{\theta_{\text{inner}} \cdot \theta_{\text{outer}}}$
δ	Composite-ray half-spread; $\delta = \frac{1}{2}\log(\theta_{\text{outer}}/\theta_{\text{inner}})$
r	Rebase factor
L_0	Position leverage, frozen at band-open
carvedNotional	Frozen carved-slice notional [USD]
carvedEntryEquity	Frozen carved-slice equity at open [USD]
entryPerpMark	Perp's mark at band-open [USD/asset]; MVP-collapsed to oracle

B Derivation of the Continuous Case

We sketch the derivation of the small-trade limit of the trade formula, which underlies the marginal-price expression used in Section 12.1.

A trade with cash leg Δy moves the reserves point along the trajectory hyperbola $(x - \alpha)(y - \beta) = \alpha\beta$. Differentiating the conservation relation with respect to a path parameter, and using $\alpha = xw$, $\beta = y(1 - w)$, the marginal exchange rate at the reserves point is

$$\text{mp} \;=\; -\frac{dy}{dx} \;=\; \frac{\alpha \cdot y^2}{\beta \cdot x^2}.$$

This is the local price of underlying in terms of cash at the reserves point. For a finite trade Δy, the proceeds are the path integral of mp along the swept arc of the trajectory hyperbola; integrating against y from y to $y' = y + \Delta y$ yields

$$\Delta x \;=\; -\,\frac{\alpha\beta \cdot \Delta y}{(y - \beta)\,(y' - \beta)},$$

which is the closed form given in the Trade Formula of Section 5. In the small-Δy limit this reduces to $\Delta x \approx -\,\text{mp} \cdot \Delta y / [\,1 - (\Delta y)/(y - \beta)\,]$, recovering mp as the limiting marginal rate.

Three derived facts follow directly. First, slippage — the gap between the proceeds of a finite trade and what they would have been at the constant pre-trade marginal price — is by construction the integrated curvature of mp along the swept arc. Second, the displacement of the reserves point along the trajectory hyperbola is bounded in y by the asymptote $y = \beta$, where the marginal price diverges; trades that would carry the reserves point past $y = \beta$ are forbidden by the closed-form expression's pole. Third, the same closed form applies for a trade at any *trade point* on the curve (Section 5); the integration is local to the arc swept by that specific transaction, and the per-strike trade-point view of Section 12.1 reduces to the same expression evaluated at the appropriate (x_T, y_T).

C Composite-Ray Identities

C.1 The within-wing identity

A spread between an inner strike θ_{inner} and an outer strike θ_{outer} (same wing) reduces to a single transaction at the composite ray

$$\theta^* = \sqrt{\theta_{\text{inner}} \cdot \theta_{\text{outer}}}, \qquad \delta = \tfrac{1}{2} \log(\theta_{\text{outer}}/\theta_{\text{inner}}).$$

The shortcut form $N \cdot \text{mark}(\theta^*) \cdot 2 \sinh |\delta|$ and the barrier-by-barrier form $N \cdot (\text{mark}(\theta_{\text{inner}}) - \text{mark}(\theta_{\text{outer}}))$ are *provably equal*, an identity rather than an approximation. The identity holds at all reachable pool states (x, y, w), on either wing.

Proof sketch.

For a call leg with $\theta < s_{\text{Norm}}$ on both endpoints, $\text{mark}(\theta) = s_{\text{Norm}}/\theta$. The barrier-by-barrier form expands to $N \cdot s_{\text{Norm}} \cdot (1/\theta_{\text{inner}} - 1/\theta_{\text{outer}})$. The shortcut form expands to $N \cdot (s_{\text{Norm}}/\theta^*) \cdot 2 \sinh \delta$. Using $\theta^* = \sqrt{\theta_{\text{inner}}\theta_{\text{outer}}}$ and $2 \sinh \delta = e^{\delta} - e^{-\delta} = \sqrt{\theta_{\text{outer}}/\theta_{\text{inner}}} - \sqrt{\theta_{\text{inner}}/\theta_{\text{outer}}}$, the second form simplifies to $N \cdot s_{\text{Norm}} \cdot (1/\theta_{\text{inner}} - 1/\theta_{\text{outer}})$. The two forms coincide. The put-wing case is symmetric.

C.2 Extension across the OTM-to-ITM boundary

The identity extends across the OTM-to-ITM boundary under the effective-strike substitution: at any moment, replace the original strike by the pool's live s_{Norm} if the leg is ITM, leaving it unchanged if OTM. Geometrically, this corresponds to the ray-parking mechanism of Section 12.1: the leg's effective ray clamps at ATM when the strike has been crossed. The substitution preserves the form of the identity and gives the leg's full closed-form value across all regimes. This boundary-extending identity is the content of verified result C1.

C.3 Cross-wing obstruction

A band — sold leg on one wing, bought leg on the other — does *not* compose to a single composite ray. A fully-OTM band has value of the form $A \cdot s_{\text{Norm}} + B/s_{\text{Norm}}$, and no single ray yields both monomials. The composite-ray shortcut is therefore valid per leg within a wing, but not across wings; a band is always two legs. The cross-wing obstruction is a separately verified result; the same wing-based distinction (rather than a transact-vs-value distinction) holds in both the transactional and the valuational use of the shortcut.

D The Three-Operator Family

The pool's state (x, y, α, β) admits exactly three primitive operators that move it. They are distinguished by their scaling signatures on the conserved quantities $\alpha = xw$ and $\beta = y(1 - w)$:

Operator	Effect on α	Effect on β
trade	$\alpha \to \alpha$	$\beta \to \beta$
rebase	$\alpha \to r \cdot \alpha$	$\beta \to \beta$
liquidity	$\alpha \to \lambda \cdot \alpha$	$\beta \to \lambda \cdot \beta$

A trade keeps both conserved quantities fixed (within-conservation move on the trajectory hyperbola). A rebase scales α by an anisotropic factor r (oracle-frame move; $x \to rx$, y unchanged). A liquidity event scales α and β together by an isotropic factor λ (proportional reserve scaling; $x \to \lambda x$, $y \to \lambda y$).

Generating-set property.

These three operators together generate the entire reachable pool-state space from any initial state. Any modification of (x, y, α, β) that arises in the protocol's execution is a finite composition of trade, rebase, and liquidity. The distinguishing (α, β)-signatures provide a uniqueness witness: any pool-state transformation can be uniquely decomposed into these three primitive moves by its action on α and β.

Commutation.

Trades and rebases commute on final state (Strike Normalisation, Section 6); they do not commute on slippage, as a rebase changes the swept arc of a subsequent trade. Trade-and-liquidity commutation is more subtle: a liquidity event applied before a trade re-scales the pool, changing the marginal price the trade sees; the order matters for proceeds, though the post-liquidity-then-trade and post-trade-then-liquidity states are related by a scaling.

E The Three-Stage Unit Chain

Settlement values are kept in three distinguishable stages, with exactly one conversion between the last two. The discipline is what makes the dimensional-check argument of the Settlement Formula section airtight at the level of the formula.

Stages.

- **Stage 1** — bare fraction. The mark function returns a number in $(0, 1]$, dimensionless. Leg-level evaluations mark(θ) live here.
- **Stage 2** — carved-perp units. A leg value $q \cdot$ mark is a fraction of the band's carved origin slice; arithmetic over these quantities (including raw_net $= Y - X$) stays in stage 2.
- **Stage 3** — dollars. Reached by a single multiplication of raw_net (stage 2) by carvedEquityAtClosure (in dollars), followed by leverage amplification.

The single-multiply discipline.

Exactly one stage-2-to-stage-3 conversion occurs per band, and it occurs in the tail multiplication on raw_net. Per-leg quantities do not individually cross stages 2-3; if they did, raw_net would be dimensionally inconsistent (subtraction across mixed-unit quantities). The single-multiply discipline is the implementation-level expression of this constraint.

Dimensional reading of the settlement formula.

With this stage discipline, the settlement formula of Section 12.1 — with its explicit / entryPerpMark normalisation in attributablePnL — balances dimensionally at every step. The attributablePnL formula's middle factor is a dimensionless fractional move ([USD/asset]/[USD/asset] $= 1$), so the formula's output is [USD] $\cdot\, 1 =$ [USD], consistent with carvedEntryEquity. Without the normalisation, the same formula would yield [USD^2/asset], an inconsistent unit. The dimensional check is, in effect, what enforces the single-multiply discipline.

F State-Space Geometry

The pool's reachable state space carries a clean geometric structure that has been characterised in supplementary verification work.

Trajectory hyperbola.

At fixed conserved (α, β), the set of reachable states under trading alone is one branch of a rectangular hyperbola with centre (α, β): the admissible arc is the open region $\{(x, y) : x > \alpha,\, y > \beta,\, (x-\alpha)(y-\beta) = \alpha\beta\}$. The arc has asymptotes $x = \alpha$ and $y = \beta$; as $x \to \alpha^+$, $y \to \infty$ (and $w \to 1^-$), and as $x \to \infty$, $y \to \beta^+$ (and $w \to 0^+$).

Tangency with the pool curve.

At every reachable reserves point, the trajectory hyperbola and the pool curve $x^w \cdot y^{1-w} = k$ are tangent — they share the same slope at that point. This tangency is why pricing read off the pool curve is faithful as a description of trade-induced motion, even though the trajectory and the pool curve are distinct objects: the pool curve provides the shape the protocol skews, the trajectory hyperbola is the path trades trace within that shape, and at the reserves point the two coincide tangentially.

Rebases as a one-parameter group.

The rebase operator with factor r forms a multiplicative one-parameter group acting on the positive quadrant of (x, y) space. Composition: $\mathrm{rebase}(r_1) \circ \mathrm{rebase}(r_2) = \mathrm{rebase}(r_1 \cdot r_2)$. Identity: $\mathrm{rebase}(1) = \mathrm{id}$. Inverse: $\mathrm{rebase}(r)^{-1} = \mathrm{rebase}(1/r)$. The orbits are horizontal rays of constant (y, w); a rebase moves the reserves point along the x-axis while preserving the other coordinates.

Joint invariant.

Under the combined action of trade and rebase, the conserved quantity $\beta = y \cdot (1-w)$ is preserved by both operations. It is the joint invariant: the orbit space of the combined action is foliated by level sets $\{\beta = \mathrm{const}\}$, with the two operations acting within each leaf. (The other conserved quantity α is preserved by trade but scaled by rebase; the invariant k is scaled by r^w under rebase, so neither α nor k is jointly invariant.) The reachable state space modulo trade-and-rebase action is therefore 1-dimensional, parametrised by β.

The trajectory hyperbola characterisation, the tangency property, the rebase group structure, and the joint-invariant identification are all formally verified in Lean 4 (see Appendix G). The deeper question of whether the state space admits a metric structure under which trades and rebases act as isometries is an open direction for which preliminary work shows rebases are isometries of a log-coordinate metric while trades are not (trades change k, which any rebase-invariant isometry would preserve); a complete characterisation is deferred.

G Formal Verification (Lean 4)

Selected core claims have been verified in Lean 4 [6] against the Mathlib standard library [10]. All results listed below are checked sorry-free and depend only on standard Mathlib axioms (propext, Classical.choice, Quot.sound); no domain-specific axioms or stubbed lemmas are used.

- **C1 — composite-ray boundary identity.** The composite-ray closed form for settlement holds across the OTM-to-ITM boundary, under the effective-strike substitution (original strike if OTM, pool's live s_{Norm} if ITM). The identity is proved as an equality of closed forms in all regime combinations; the substitution preserves the form's value across the OTM-to-ITM transition. This establishes the close-equals-exercise equivalence as a proven property.
- **C4 — no-internal-arbitrage biconditional.** The costless-collar capital-efficiency surplus $\text{collarSurplus}(\theta, w) = 0$ for all admissible strikes θ if and only if the pool is symmetric, $w = \frac{1}{2}$. A genuine biconditional, derived from the concrete $\min(\text{slope}, 1/\text{slope})$ mark formula with no domain-specific axioms; the proof includes an explicit skew counterexample establishing the negative direction.
- **CompositeRayBand — cross-wing obstruction.** A band's two-leg opposite-wing structure does not reduce to a single composite ray. The proof exhibits the $A \cdot s_{\text{Norm}} + B/s_{\text{Norm}}$ form of a fully-OTM band's value and verifies that no choice of single (θ, q) reproduces it. Confirms that the within-wing composite-ray shortcut is the correct scope, not the cross-wing case.
- **Tangency and rebase composition.** The trajectory hyperbola $(x - \alpha)(y - \beta) = \alpha\beta$ is tangent to the pool curve $x^w \cdot y^{1-w} = k$ at every reachable reserves point. Trades and rebases commute on final state. A rebase multiplies k by r^w. Each statement is proved as a clean identity via real-analytic algebra.
- **State-space geometry.** The trajectory hyperbola is fully characterised as a rectangular hyperbola with centre (α, β) and admissible arc $\{x > \alpha,\, y > \beta\}$; both endpoint limits ($y \to \infty$ as $x \to \alpha^+$, and $y \to \beta^+$ as $x \to \infty$) are formally verified as Filter.Tendsto statements. Rebases form a one-parameter multiplicative group (composition, identity, inverse all proved). The conserved quantity β is the joint invariant under combined trade-and-rebase action. The geometric grid structure (trajectory curves $\times$ rebase orbits) at the reserves point is established formally.

Verification artifacts are available from the authors on request, and will be deposited in a public repository upon publication [5].

H Worked Settlement Example

A small numerical example may help anchor the Settlement formulas. Consider a trader holding a long perpetual future on an underlying with mark \$100 at the moment they open a band. The band is carved against a slice of size \$10,000 notional and \$1,000 entry equity (perp leverage $10\times$), with band leverage $L_0 = 2$.

A short while later, the perp's mark has moved to \$110, and the trader closes the band. The OTM leg reverses on the AMM and produces a stage-2 value of 0.30 (carved-perp units);

the ITM leg settles to cash with stage-2 value 0.05. Then:

$$\begin{aligned}
\text{raw_net} &= 0.30 - 0.05 = 0.25 \quad \text{(carved-perp units)} \\
\text{attributablePnL} &= \$10{,}000 \cdot (110 - 100)/100 \;=\; \$1{,}000 \\
\text{carvedEquityAtClosure} &= \$1{,}000 + \$1{,}000 \;=\; \$2{,}000 \\
\text{trader_payout} &= 2 \cdot 0.25 \cdot \$2{,}000 \;=\; \$1{,}000 \\
\text{club_delta} &= 1 \cdot 0.25 \cdot \$2{,}000 \;=\; \$500
\end{aligned}$$

The trader receives \$1,000 from the band and the counterparty club balance changes by \$500. Note the asymmetry between the two: $L_0 = 2$ means the trader keeps twice the per-band net while the counterparty bears the unleveraged difference, with the equity floor ensuring the protocol does not pay out beyond what the counterparty club can absorb.

This worked example also illustrates the dimensional discipline of Appendix E: raw_net is dimensionless (0.25 has no unit); carvedEquityAtClosure is in dollars; the product $0.25 \cdot \$2{,}000$ converts stage-2 to stage-3 in a single multiplication; L_0 is dimensionless and amplifies. No formula step crosses dimensions illegitimately.